

15 Ideal gases

A · 2026 examination · Concept guide and original exam-style practice

Understand the concept

The ideal gas equation $pV = nRT$ links macroscopic state variables. In kinetic theory, pressure arises from collisions and mean translational kinetic energy per molecule is $3kT/2$. Use kelvin, pascals and cubic metres; a real gas approaches ideal behaviour at low pressure and high temperature.

Worked example: follow the reasoning

For 0.20 mol at 300 K in 0.0050 m³, $p = nRT/V = 0.20 \times 8.31 \times 300/0.0050 \approx 1.0 \times 10^5$ Pa.

Quick check · multiple choice

A Which is the ideal-gas equation?

[1]

A $p/T = nRV$

B $pV = nR/T$

C $pV = nRT$

D $p = nVRT$

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A sealed rigid vessel at 300 K has $p = 100$ kPa. Find p at 450 K and state the assumption.

[3]

Final answer: _____

Answers and explanation

Quick check: C. $pV=nRT$

Question 1 [3 marks]

At fixed volume and amount, p/T is constant [1]. $p_2 = 100 \times 450/300 = 150 \text{ kPa}$ [1]. Assume ideal gas behaviour [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Physics 9702, 2025–2027. <https://www.cambridgeinternational.org/Images/664565-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.